ESP32-S3 Module 2: Water Quality & Flow Hub

PCB Hardware Engineering & Layout Specification

Doc ID: SPEC-PCB-M2-WQ-V1.0
Target MCU: ESP32-S3-WROOM-1 / DevKitC-1
Date: October 2026 | **Status:** Production Ready

MAIN DC POWER INPUT	LOGIC LEVELS	ANALOG CHANNELS	DISPLAY BUS
5.0V DC (or 12V Buck)	**3.3V LVTTL / 5V Tolerant**	**2x 12-Bit ADC (TDS, pH)**	**I2C (SDA: 17, SCL: 18)**

1. MASTER MCU PIN MAPPING & SCHEMATIC NETLIST

ESP32 PIN	SCHEMATIC NET	SUBSYSTEM DEVICE	SIGNAL TYPE	RAIL	SIGNAL CONDITIONING & PASSIVES
GPIO 1	NET_TDS_ADC	Analog TDS Meter	Analog (ADC1_CH0)	3.3V	RC Filter: 100Ω series + 100nF MLCC to GND
GPIO 2	NET_PH_ADC	pH-4502C Sensor (Po)	Analog (ADC1_CH1)	5.0V*	RC Filter: 100Ω series + 100nF MLCC to GND
GPIO 4	NET_DHT11_DATA	DHT11 Air Sensor	Single-Wire GPIO	3.3V	Pull-up: 4.7kΩ to 3.3V bus
GPIO 5	NET_FLOW_PULSE	FS200A Flow Sensor	Ext Interrupt	5.0V/3V3	10kΩ pull-up to 3.3V + 1kΩ series limit + 1nF cap
GPIO 17	NET_I2C_SDA	2004 LCD Display	I2C0 Data (Open-Drain)	3.3V	Pull-up footprint: 4.7kΩ to 3.3V
GPIO 18	NET_I2C_SCL	2004 LCD Display	I2C0 Clock (Open-Drain)	3.3V	Pull-up footprint: 4.7kΩ to 3.3V

CRITICAL HARDWARE RULE (Octal Flash / PSRAM Restriction): On the ESP32-S3, **GPIO 33, 34, 35, 36, and 37 are strictly reserved** for the internal high-speed Octal SPI Flash/PSRAM interconnect. Routing any sensor or header trace to GPIO 33–37 will permanently crash the microcontroller bootloader.

2. HEADER PINOUTS & PERIPHERAL CONNECTORS (FOOTPRINT SPECS)

J1 — Screw Term 5.08mm

Main DC Power Input

1	+12V / +5V EXT	Power In
2	GND	0V Return

J2 — JST-XH 2.54mm (3P)

DHT11 Temp & Humidity

1	+3V3	3.3V VCC
2	NET_DHT11_DATA	GPIO 4
3	GND	Ground

J3 — JST-XH 2.54mm (3P)

Analog TDS Meter

1	NET_TDS_ADC	GPIO 1
2	+3V3	3.3V VCC
3	GND	Ground

J4 — Screw / JST-XH (3P)

FS200A Flow Sensor

1	+5V0 / +3V3	Red Wire
2	NET_FLOW_PULSE	GPIO 5
3	GND	Black Wire

J5 — JST-XH 2.54mm (4P)

pH-4502C Probe Module

1	+5V0	Mandatory 5V
2	GND (Power)	Ground
3	NET_PH_ADC (Po)	GPIO 2
4	GND (Analog)	Ground

J6 — 2.54mm Header (4P)

2004 I2C LCD Display

1	GND	Ground
2	+5V0	Mandatory 5V
3	NET_I2C_SDA	GPIO 17
4	NET_I2C_SCL	GPIO 18

3. ANALOG FRONT-END CONDITIONING & NOISE SUPPRESSION

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            ANALOG INPUT LOW-PASS FILTER (GPIO 1 & GPIO 2)
Sensor Signal >---+------[ R_FILTER: 100 Ohm ]------+------> ESP32-S3 ADC Pin
                  |                                 |
               [ TVS ]                         [ C_FILTER ]
               SMAJ3.3A                        100nF MLCC X7R
                  |                                 |
                GND                               GND
* Function: Eliminates 50Hz/60Hz AC mains hum and submersible pump motor inductive spikes from water tanks.
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4. PCB LAYOUT & ROUTING ENGINEERING GUIDELINES

Trace Geometries & Power Distribution

- **+5V0 & +3V3 Power Traces:** 24 to 32 mils (0.6mm – 0.8mm) to minimize I^2R drop.
- **Analog Signals (GPIO 1, GPIO 2):** 10 to 12 mils. Keep strictly away from relay coils, switching DC-DC inductors, and crystal oscillators.
- **Digital Signals (I2C, DHT, Flow):** 10 to 12 mils with 3W spacing to prevent crosstalk.

Ground Plane & Decoupling Strategy

- **Solid Ground Pour:** Maintain an unbroken GND copper plane on Layer 2 (Bottom layer).
- **Decoupling Capacitors:** Place 100nF ceramic caps within **< 3mm** of sensor connector pins and ESP32 VDD pins.
- **Star Grounding:** Connect analog sensor grounds (TDS & pH) directly to main system GND to prevent return current offsets.

5. RECOMMENDED BILL OF MATERIALS (BOM)

DESIGNATOR	VALUE / PART	PACKAGE / FOOTPRINT	DESCRIPTION & PURPOSE
U1	ESP32-S3-DevKitC-1	2x22 Pin 2.54mm Socket	Dual-core 240MHz MCU with Wi-Fi & BLE
R1, R2	100 Ω ±1%	0805 SMD / 0603	ADC Low-Pass Filter Series Resistors (GPIO 1 & 2)
C1, C2	100 nF 50V X7R	0805 SMD / 0603	ADC Low-Pass Filter Shunt Capacitors
R3, R4	4.7 kΩ ±5%	0805 SMD	I2C Bus Pull-Ups to +3V3 (SDA & SCL)
R5	4.7 kΩ ±5%	0805 SMD	DHT11 Single-Wire Data Pull-Up to +3V3
R6, R7	10 kΩ / 1 kΩ	0805 SMD	Flow Sensor Pull-Up (10k) and Series Limit (1k)
C3, C4	100 µF 25V / 10 µF	Radial Electrolytic / 1206	Bulk Power Rail Decoupling (+5V & +3.3V)
D1	SS34 (Schottky)	SMA / DO-214AC	Reverse Polarity Protection Diode on Main DC In